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$$\begin{aligned}w &= \frac{1}{u^2 + v^2}, u = \cos 2t, v = \sin 2t \\ \frac{\partial w}{\partial u} &= -\frac{2u}{(u^2 + v^2)^2}, \frac{\partial w}{\partial v} = -\frac{2v}{(u^2 + v^2)^2} \\ \frac{du}{dt} &= -2 \sin 2t, \frac{dv}{dt} = 2 \cos 2t \\ \frac{dw}{dt} &= \frac{\partial w}{\partial u} \frac{du}{dt} + \frac{\partial w}{\partial v} \frac{dv}{dt} \\ &= \frac{4u \sin 2t - 4v \cos 2t}{(u^2 + v^2)^2} \\ &= 4 \cdot (\cos 2t \sin 2t - \sin 2t \cos 2t) = 0\end{aligned}$$

References